

Engineering Project Management

ET 8401

Instructor: Kadigi (*Ph.D.*)

Email: ibrahim.kadigi@sua.ac.tz

0752-887-661

Why this course to you?

- In this course, you will learn about the main issues that differentiate the project management of ICT initiatives from other kind of projects.

COURSE OUTLINE

Course Contents

- ICT Project Failure and Success
- Project Management Foundation
- Project Management Initiation
- Project Scope Definition
- Time Management and Scheduling
- Resource Planning
- Cost Management and Control
- Communication Management
- Project Risk Management
- Procurement and Sourcing
- Controlling and Managing Change
- Quality Assurance and Control
- Phase and project closure
- Network Based Techniques

Course Contents

ICT Project Failure and Success

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- Reasons for ICT project Failure
- Reasons for ICT project success
- ICT Projects – what makes them different

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Course Contents ...

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Introduction to ICT Project Management

Introduction

What is Project Management?

Introduction ...

- Project Management is defined as a sequence of activities having one purpose that must be completed
 - ❖ by a specific time
 - ❖ within budget and
 - ❖ according to expected requirements

Introduction ...

Primary objective of PM

- ❑ The primary objective of project management is to ensure that a project is completed successfully within the defined scope, on time, and within budget while meeting or exceeding the stakeholders' expectations.
- ❑ This involves balancing the project constraints, which are often represented by scope, time, cost, quality, resources, and risk.

Introduction ...

Primary objective of PM

- ❑ **Scope:** Clearly defining and managing what is included and what is excluded from the project to meet the specific objectives laid out at the start.
- ❑ **Time:** Delivering the project's deliverables on schedule by effectively managing the time allocated for each activity and ensuring timely completion.
- ❑ **Cost:** Completing the project within the approved budget, managing finances efficiently throughout the project lifecycle to maximize resource allocation.

Introduction ...

Primary objective of PM

- ❑ **Quality:** Ensuring that the project's outputs meet the necessary quality standards and stakeholder expectations, aligning with the project's objectives.
- ❑ **Resources:** Efficiently and effectively deploying project resources, including people, technology, and materials, to achieve the project goals.
- ❑ **Risk:** Identifying potential risks early on and managing them proactively to minimize their impact on the project.

Introduction ...

- ICT Project Management is a specialized area of project management that requires precise oversight over complex interactions between software, hardware, network infrastructure, and data.
- It differs from general project management due to its highly technical elements and rapid pace of change in technology.
- ICT project management typically follows the traditional phases of project management but with a focus on technology-specific considerations.
- These phases include initiation, planning, execution, monitoring and controlling, and closure

Introduction...

Some examples of ICT projects are listed as follows:

❖ **Information Systems /Portals**

- e-Government portals
- City websites
- Core banking information system
- Immigration passport system

❖ **Network and Infrastructure**

- Implementation of network (infrastructure) and equipment (routers, switches, etc)
- Microsoft exchange email system

❖ **ICT Consultancy and Services**

- e-Government ICT HR master plan
- Network design and consultancy study
- ICT master plan and e-Government roadmap.

Introduction...

Success and Failure ICT Related Projects

- ICT projects, like any complex undertakings, can either succeed spectacularly or fail disastrously.
- The reasons for success and failure in ICT projects can be numerous and varied, often intertwined with both technical and managerial factors.
- Understanding these reasons can help you or your organization better prepare and execute your ICT initiatives.

Introduction...

Reasons for ICT Project Success

❖ Clear Goals and Objectives

- Successful ICT projects have well-defined, realistic goals that align with business objectives, providing a clear roadmap and helping to keep the project on track.

❖ Strong Project Management

- Effective leadership and management, including meticulous planning, risk management, and resource allocation, are crucial.
- This also includes applying appropriate project management methodologies (like Agile or Waterfall, depending on the project needs).

Introduction...

Reasons for ICT Project Success ...

❖ Stakeholder Engagement

- Regular communication with and active involvement of all stakeholders, including end-users, ensures that the project meets the actual needs of the organization and receives the necessary support.

❖ Competent Project Team

- Having a skilled and experienced team that can address the technical, management, and operational aspects of the project is vital.
- Continuous training and development are also key.

Introduction...

Reasons for ICT Project Success ...

❖ Adherence to Budget and Schedule

- Successful projects stay within budget and meet their deadlines, which is often a result of effective scope and change management to prevent scope creep.

❖ Quality Control Processes

- Implementing robust quality assurance and testing processes ensures that the project meets the required standards and performs as expected.

❖ Effective Change Management

- Successfully managing the change that comes with new ICT implementations helps organizations smoothly transition to new systems and processes.

Introduction...

Reasons for ICT Project Failure

❖ Unclear or Shifting Project Objectives

- Lack of clear goals or changing objectives without proper adjustments can derail a project from meeting its intended outcomes.

❖ Poor Project Management

- Inadequate planning, underestimating the complexity of the project, poor risk management, and ineffective leadership can all lead to project failure.

❖ Lack of Stakeholder Engagement

- Insufficient involvement of end-users and other stakeholders can lead to a lack of buy-in and support, resulting in a product that does not meet the needs of its users.

Introduction...

Reasons for ICT Project Failure

❖ Inadequate Resources

- This can include insufficient budget, time constraints, or lack of necessary skills within the project team, all of which can impede project success.

❖ Scope Creep

- Uncontrolled changes or continuous growth in project scope without adjustments to time, cost, and resources can overwhelm the project team and lead to project failure.

Introduction...

Reasons for ICT Project Failure

❖ Technical Issues

- Underestimating technical challenges, choosing the wrong technologies, poor system integration, or inadequate testing can cause significant problems.

❖ Resistance to Change

- If the project does not address the change management adequately, users may resist adopting the new solution, thereby undermining the project's success.

Introduction...

How to Mitigate Risk and Enhance Project Success

❖ Comprehensive Planning and Analysis

- Spend adequate time during the project initiation phase to plan thoroughly and perform detailed analysis.

❖ Continual Monitoring and Evaluation

- Regularly check project progress against benchmarks and adapt strategies as needed.

❖ Communication and Documentation

- Maintain open lines of communication across all levels of the project and document all decisions and changes thoroughly.

❖ Invest in Training

- Equip your team and the end-users with the necessary knowledge and skills to handle the new systems effectively.

Project Management Foundation

Project Management Foundation

- Project management is a critical discipline used across various industries to plan, execute, control, and close projects.
- It involves the application of knowledge, skills, tools, and techniques to project activities to meet the project requirements.
- Understanding the foundation of project management provides a base for effective project execution and delivery.

Project Management Foundation

Foundational Elements of Project Management

➤ 1. Project Management Framework

- Projects typically follow a sequential process, including initiation, planning, execution, monitoring and controlling, and closing (**project life cycle or phases**).

➤ 2. Core Functions

- Ensuring the project includes all the work required, and only the work required, to complete the project successfully (**scope Management**).
- Developing and managing a timetable that ensures the project is completed on time (**Schedule and Time Management**).
- Planning and managing the budget to ensure the project is completed within the approved expenditure (**cost management**).

Project Management Foundation

Foundational Elements of Project Management

➤ 2. Core Functions....

- Determining quality policies, objectives, and responsibilities so that the project will satisfy the needs for which it was undertaken (*quality management*).
- Identifying, acquiring, and managing resources needed for the successful completion of the project (*resource management*).
- Acquiring or procuring goods and services from outside sources to meet project requirements (*Procurement Management*).
- Engaging all stakeholders appropriately, and managing their expectations and satisfaction (*stakeholder management*).

Project Management Foundation

➤ 3. Project Composition and Responsibilities

- The composition required to carry out the project management shall consist of the following parties:
 - Project steering committee
 - Project Manager
 - Project Team

Project Management Foundation

➤ 3. Project Composition and Responsibilities ...

- **Project Steering Committee**

- Management group chaired by a senior executive as the Project Sponsor

Main Responsibilities of the Project Steering Committee

- Provide management guidance and feedback
- Provide access to resources (personnel, information, etc.)
- Review all recommendations on opportunities, approaches and resources
- Maintain focus on business needs
- Ensure adequate funding is available to support the project
- Make decisions concerning resource conflicts.
- Ensure risk assessment is regularly reviewed and the risks are managed

Project Management Foundation

➤ 3. Project Composition and Responsibilities ...

• Project Manager

- A person assigned by the Project Steering Committee for day-to-day management on their behalf

Main Responsibilities of members of the Project Manager

- Select the project team, together with the sponsor
- Maintaining a close working relationship with the sponsor
- Identify and manage the project stakeholders
- Identify and manage the risks
- Allocate and secure resource commitments
- Monitoring and tracking the project progress
- Solving problems that interfere with progress
- Controlling cost
- Leading the project team
- Informing stakeholders of progress
- Delivering the project deliverables on time
- Manage the performance of the project team

Project Management Foundation

➤ 3. Project Composition and Responsibilities ...

- **Project Team**

- Individuals selected to achieve the deliverables

Main Responsibilities of the Project Team

- Helping the Project Manager to deliver the objectives
- Participate in team meetings, visits, and workshops
- Use their competencies to carry out the tasks they have been given
- Alerting the Project Manager to any risks that appear
- Provide information for the project documentation as needed

Project Management Foundation

➤ 4. Project Management Tools

- **Project Management Software**

- Tools like Microsoft Project, Asana, and Trello help in scheduling, resource allocation, and budget management.

- **Gantt Charts**

- Visual representations of the project schedule where activities, along with their start and end dates, are shown against a calendar.

- **Critical Path Method**

- A step-by-step project management technique to identify activities on the critical path.
- It is an approach to project scheduling that determines the longest sequence of dependent tasks and gauges the minimum amount of time needed to complete them.

- **Earned Value Management**

- A technique that combines scope, schedule, and resource measurements to assess project performance and progress.

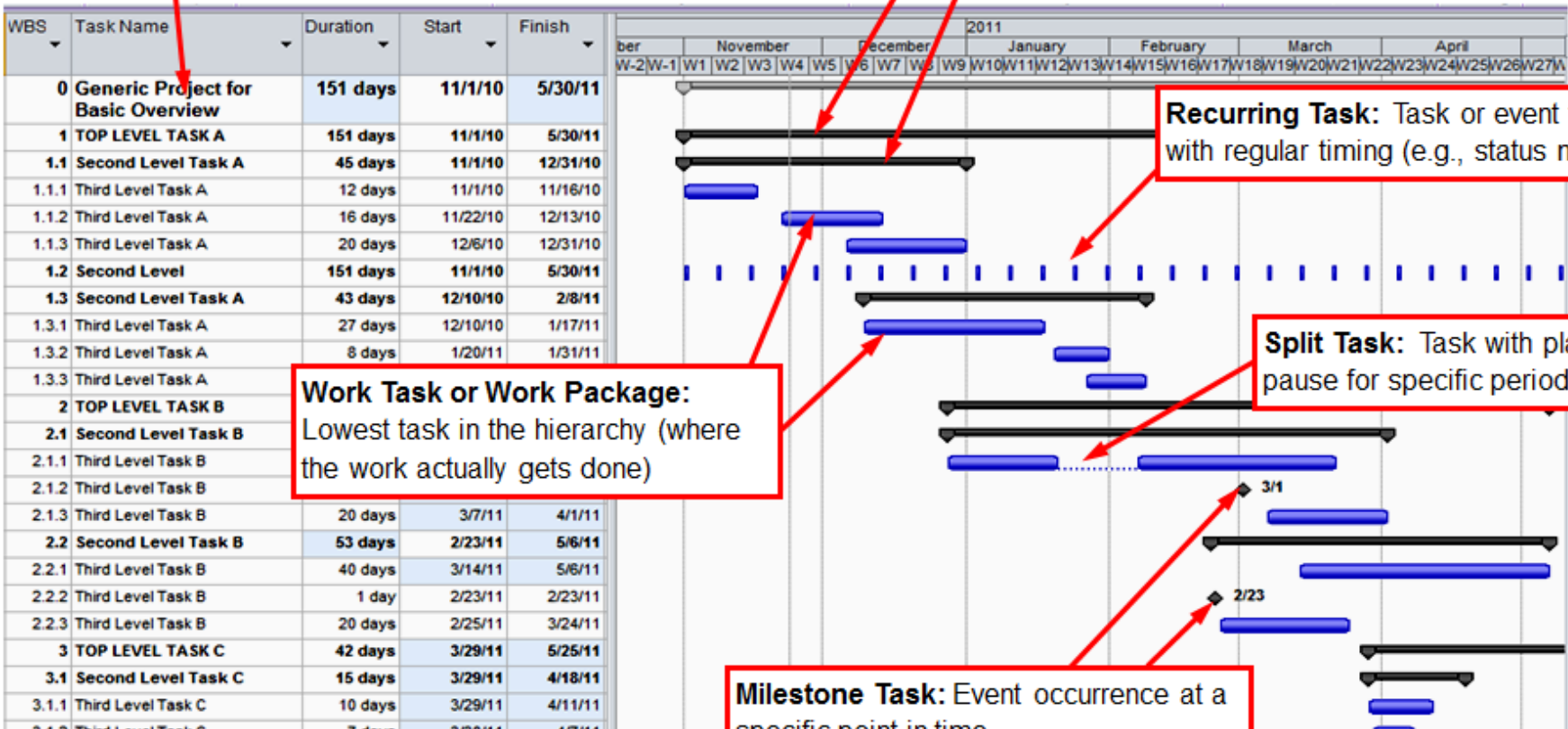
Gantt Charts

	Activity	Hours needed	Timeline [h]																																							
N			Work day 1								Work day 2								Work day 3								Work day 4															
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32								
1	3D CAD modelling of the assembly	10																																								
2	Assembly drawing and import in Unity 3D as trackable	6																																								
3	3D CAD model alignment with natural feature	1																																								
4	Updating BOM and functionalities linked to its components	2																																								
5	Baloons alignment with natural feature	1																																								
6	Highlight alignment with natural feature	3																																								
7	Animations of the assembly	1																																								
8	Creation of component recognition labels and scripts	2																																								
9	Updating link between AR and VR scenes	1																																								

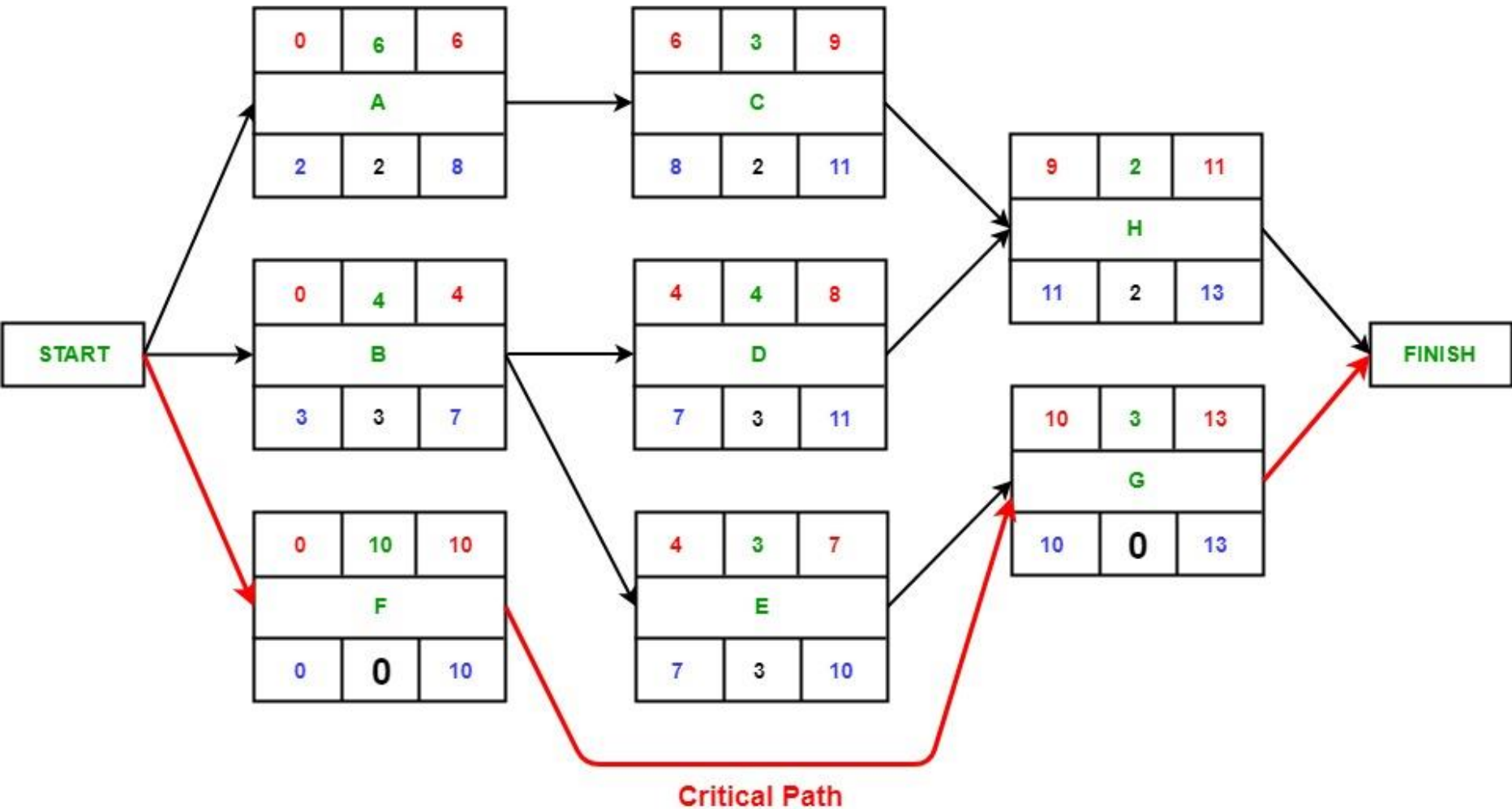
Gantt Charts

Project Summary Task: Special task identifying entire project

Summary Task: Any with lower-level subtasks



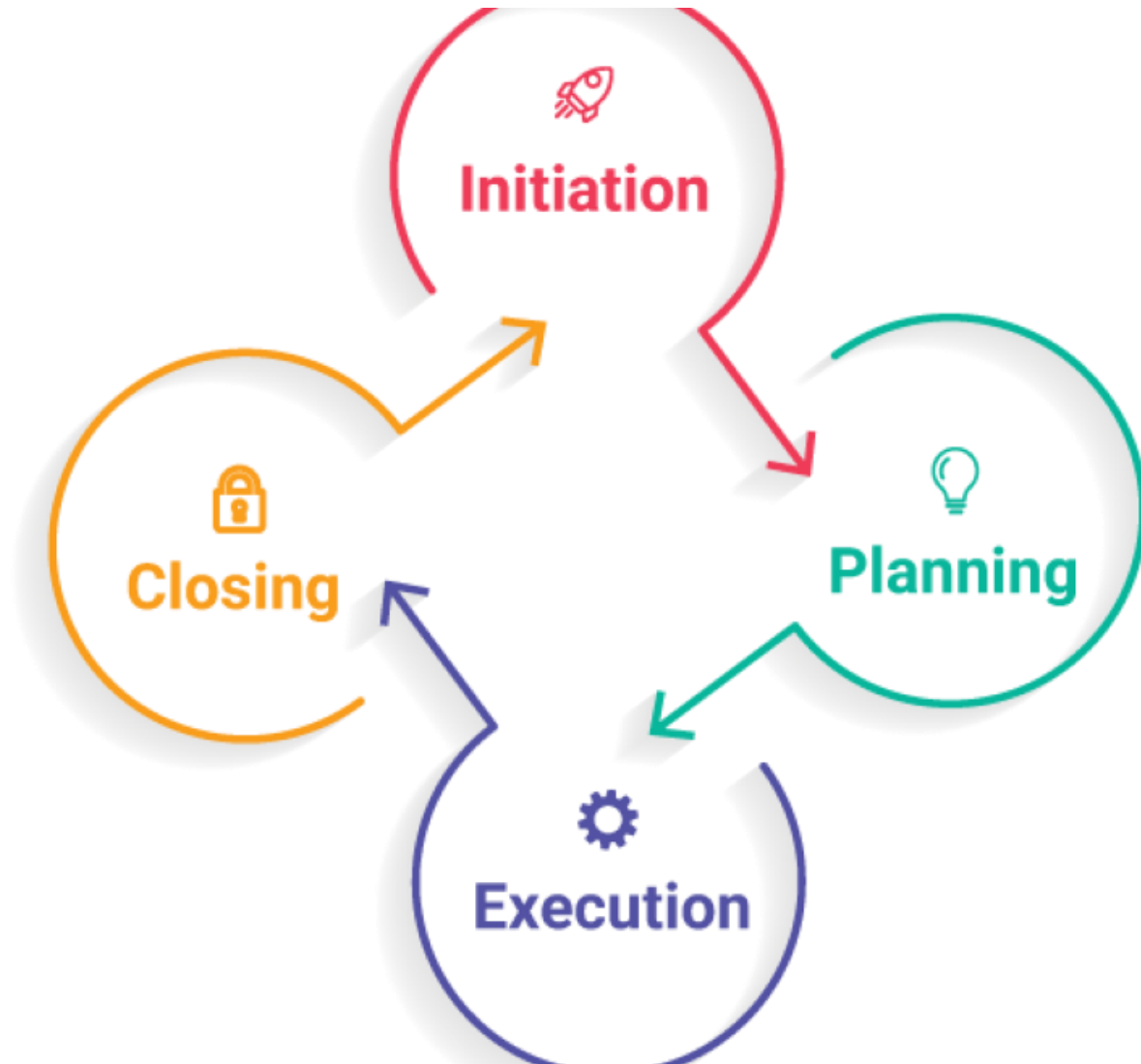
Critical Path



Project Life Cycle

Project Life Cycle

- The project life cycle (PLC) is a fundamental concept in project management that describes the stages through which a project progresses from **start** to **finish**.
- It provides a structured approach to planning and executing projects, ensuring consistency, clarity, and control throughout the entire process.
- The life cycle typically comprises four primary phases: **initiation**, **planning**, **execution**, and **closure**.
- Each phase has specific objectives and tasks that contribute to the overall success of the project.



Project Life Cycle

Why is the Project Life Cycle important?

- Understanding and following the project life cycle is crucial for project managers and teams because it provides:
 - A structured approach that helps to manage large and complex projects by breaking them into manageable phases.
 - Efficiency through planned stages, resources and efforts are directed appropriately, reducing waste and increasing productivity.
 - Improved control because each phase has its own process and deliverables, making it easier to monitor progress, make adjustments and manage risks effectively.

Initiation Phase

PLC Initiation Phace

Initiation Phase

- The initiation phase is the starting point of a project.
- During this phase, the project's feasibility and relevance are evaluated to ensure that it should proceed.
- This phase involves defining the project at a broad level.
- It often starts with a business case that explains the need for the project and its expected benefits.
- A feasibility study may be conducted to assess the practicality and value of the project.
- Key project roles such as the project sponsor, project manager, and other critical stakeholders are identified.

PLC – Initiation Phase

Initiation Phase Key Activities

Identifying the Project's Purpose and Scope

- This involves defining what the project is supposed to accomplish and the boundaries within which it must operate.
- The Project Objective Statement (POS) describes in concrete terms what the project aims to achieve.
- It articulates the final outcome of the project and how the success of the project will be measured.
- **Project Scope** is the boundary that defines where the project begins and where it ends.

PLC – Initiation Phase

Initiation Phase Key Activities ...

Developing the Project Charter

- This document formally authorizes the existence of the project and provides the project manager with the authority to apply organizational resources to project activities.

Stakeholder Identification

- Recognizing all parties affected by the project and understanding their influences and interests relative to the project's objectives.

PLC – Initiation Phase

Example of Initiation Phase

- Let's consider an example of the initiation phase in the context of launching a new **software development project for a mobile banking application for CRDB Bank.**
- This example will illustrate the typical activities and decisions made during the initiation phase of a project.

PLC – Initiation Phase

Example of Initiation Phase

Project: Mobile Banking Application Development

Background

A regional bank wants to develop a mobile banking application to expand its services and provide customers with a convenient, secure, and reliable mobile banking experience. The bank aims to increase customer satisfaction and compete with other financial institutions that offer advanced digital banking solutions.

PLC – Initiation Phase

Example of Initiation Phase ...

Step 1: Project Ideation and Proposal

Business Case

A business case is developed by the CRDB bank's digital services department outlining the need for a mobile banking app to enhance customer engagement, reduce operational costs, and increase revenue through new digital products.

Feasibility Study

A preliminary study assesses the technical feasibility, market demand, and financial justification for the project.

This study includes an analysis of the bank's current IT infrastructure, potential technology partners, and a cost-benefit analysis.

PLC – Initiation Phase

Example of Initiation Phase ...

Step 2: Stakeholders Identification

Internal Stakeholders

Include the bank's executive team, IT department, customer service, marketing, and compliance teams.

External Stakeholders

Potentially involve software development firms, cybersecurity consultants, regulatory bodies, and third-party service providers (e.g., payment processors).

PLC – Initiation Phase

Example of Initiation Phase ...

Step 3: Define Project Objective

Objectives

Develop a user-friendly mobile application that allows customers to perform transactions, check balances, apply for loans, and receive notifications about their accounts.

Scope

The project will include app design, development, integration with existing banking systems, compliance checks, user testing, and initial rollout in selected regions.

PLC – Initiation Phase

Example of Initiation Phase ...

Step 4: Project Charter Development

Document Creation

A project charter is prepared, detailing the project's purpose, objectives, scope, key deliverables, milestones, high-level timeline, and budget. It also outlines the roles and responsibilities of the project team.

Authorization

The project charter is reviewed and approved by the bank's board of directors, granting formal authorization to proceed with the project.

PLC – Initiation Phase

Example of Initiation Phase ...

Step 5: Appoint a Project Manager and Form a Project Team

Project Manager Appointment

A project manager with experience in digital projects and software development is appointed.

Team Formation

The project team is assembled, including internal IT staff, software developers, user experience (UX) designers, and representatives from marketing and compliance.

PLC – Initiation Phase

Example of Initiation Phase ...

Step 6: Initial Stakeholders Meeting (Kickoff Meeting)

Meeting Goals

To align the vision, discuss the project charter, and ensure all stakeholders understand their roles and the project's expectations.

Communication plan

Establishing how communication will be managed throughout the project, including reporting formats, meeting schedules, and escalation procedures.

PLC – Initiation Phase

Example of Initiation Phase ...

Concluding Remarks of Initiation Phase:

By the end of the initiation phase, the project has a clear direction, with a defined scope and objectives, and a committed team ready to move into the planning phase.

This sets the groundwork for detailed planning and execution, ensuring everyone involved has a unified understanding of the project's goals and expectations.

Planning Phase

PLC – Planning Phase

Planning phase

- The planning phase involves detailing the path from project start to completion.
- It focuses on developing a plan to guide the team through the execution and closure phases of the project.
- This includes developing project management plans for scope, schedule, cost, quality, resources, communications, risk, procurement, and stakeholder engagement.
- Setting objectives and key performance indicators (KPIs)
 - Specific, measurable, achievable, relevant, and time-bound objectives are set to ensure that project goals are clearly understood and measurable.
- Identifying potential risks and devising strategies to mitigate them is crucial for preventing project disruption

PLC – Planning Phase

An Example of a Planning Phase (continue with our previous example)

Step 1: Project Management Plan Development

Comprehensive Planning

The project manager leads the development of a comprehensive project management plan, which includes integrated plans for scope, schedule, costs, quality, resources, communication, risk, procurement, and stakeholder engagement.

Scope Management Plan

This document outlines how the project scope will be defined, validated, and controlled. It ensures that all required work (and only the required work) is included in the project.

Schedule Management Plan

Defines the project timeline, key milestones, dependencies, and the critical path. Tools like Gantt charts or project management software might be used to visualize the timeline.

Cost Management Plan

Includes budget estimates, funding allocations, and cost control measures to keep the project within the approved budget.

PLC – Planning Phase

An Example of a Planning Phase

Step 1: Project Management Plan Development

Quality Management Plan

Establishes quality benchmarks, testing protocols, and review processes to ensure the app meets all technical requirements and user expectations.

Resource Management Plan

Details the human, technological, and physical resources needed for the project, including how they will be acquired, allocated, and managed.

Communication Management Plan

Specifies the communication methods, frequencies, and protocols for updating stakeholders and managing expectations throughout the project.

PLC – Planning Phase

An Example of a Planning Phase

Step 1: Project Management Plan Development

Risk Management Plan

Identifies potential risks, assesses their impact and likelihood, and outlines mitigation strategies to manage risks proactively.

Procurement Management Plan

If external vendors or contractors are needed, this plan outlines how they will be selected and managed.

Communication Management Plan

Identifies all project stakeholders and outlines strategies for engaging them effectively throughout the project lifecycle.

PLC – Planning Phase

An Example of a Planning Phase ...

Step 2: Define Project Milestones and Deliverables

Milestones

Establish key milestones such as completion of app design, end of coding phases, testing stages, and app launch.

Deliverables

Define specific outputs for each phase, such as design prototypes, beta versions of the app, final app builds, and user manuals.

PLC – Planning Phase

An Example of a Planning Phase

Step 3: Assign Roles and Responsibilities

Team Roles

Clearly define the roles and responsibilities of each team member to ensure accountability.

This includes delineating who is responsible for design, development, testing, and deployment.

Vendor and Stakeholder Roles

Outline the responsibilities of any third-party vendors or external stakeholders, including what deliverables they are responsible for.

PLC – Planning Phase

An Example of a Planning Phase

Step 4: Risk Assessment and Mitigation Strategies

Risk Identification

Conduct workshops and meetings to identify potential risks.

Common risks might include technical challenges, delays in delivery, security vulnerabilities, and regulatory compliance issues.

Mitigation Strategies

Develop strategies to address these risks, such as adopting agile development methodologies, conducting regular security audits, and ensuring compliance with financial regulations.

PLC – Planning Phase

An Example of a Planning Phase

Step 5: Establish Budget and Funding Requirements

Budgeting

Create a detailed budget that includes estimates for labor, software, hardware, third-party services, and contingency funds.

Funding Approval

Present the budget to stakeholders for approval and secure the necessary funding to move forward.

PLC – Planning Phase

An Example of a Planning Phase

Step 6: Approval of Project Management Plan

Stakeholders Review

Present the project management plan to key stakeholders for review.

Feedback and Revisions

Incorporate feedback and make necessary revisions to the plan.

Final Approval

Obtain formal approval to proceed with the execution phase.

PLC – Planning Phase

Concluding Remarks of Planning Phase

The planning phase culminates in a detailed project management plan that acts as a blueprint for project execution.

This plan ensures all team members and stakeholders are on the same page and fully understand their roles, the project timelines, the scope of work, and the measures in place to manage risks and communications.

With this solid plan, the project is well-prepared to move into the execution phase, where the actual development and building of the mobile banking application will begin.

Execution Phase

PLC – Execution Phase

Execution Phase

- The execution phase is where most of the work on the project happens—it's about delivering on the commitments made in the project plan.
- Ensuring that human, financial, and physical resources are secured and managed effectively.
- The tasks identified during the planning phase are carried out, monitored, and controlled based on performance against the project management plan.
- Ensuring that project deliverables meet the quality standards set out in the planning phase.
- Keeping stakeholders engaged and informed throughout the project lifecycle.

PLC – Execution Phase

An Example of an Execution Phase

Execution Phase: Mobile Banking Application Development Project

Step 1: Kick-off Execution

Kick-off Meeting

Conduct a formal kick-off meeting with the project team, stakeholders, and any third-party vendors to mark the start of project execution. Reiterate the project goals, timeline, key milestones, and individual roles and responsibilities.

Step 2: Develop and Integrate

App Development

The development team starts coding according to the design specifications.

They work in iterations, allowing for incremental building and testing of the app's features.

Integration

Simultaneously, the IT team works on integrating the app with existing bank systems, such as databases for customer data, transaction systems, and authentication servers.

PLC – Execution Phase

An Example of an Execution Phase

Step 3: Quality Assurance and Testing

Unit Testing

Developers perform unit testing to ensure that individual components of the app work correctly.

Integration Testing

Test integration points between the app and existing bank systems

User Acceptance Testing (UAT)

Engage a group of beta testers or actual end-users to test the app and provide feedback. This helps ensure that the app meets user requirements and is user-friendly.

Security and Compliance Testing

Conduct thorough security checks and compliance verifications to ensure the app meets all relevant standards and regulations.

PLC – Execution Phase

An Example of an Execution Phase

Step 4: Performance Management

Monitoring and Controlling

Use project management tools to track progress against the project plan. Regular status meetings help monitor milestones, manage resources, adjust timelines, and resolve issues.

Quality Control

Implement quality control processes to adhere to the quality management plan. Regular reviews and audits ensure deliverables meet the quality standards established in the planning phase

PLC – Execution Phase

An Example of an Execution Phase

Step 5: Stakeholder Communication and Reporting

Regular Updates

Provide stakeholders with regular updates on progress, challenges, and any changes. Use the communication plan as a guide for frequency and methods of communication.

Issues Management

Record and manage any issues that arise, ensuring they are addressed promptly and do not affect the project timeline or deliverables.

PLC – Execution Phase

An Example of an Execution Phase

Step 6: Documentation

Technical Documentation

Maintain comprehensive documentation of the app's architecture, code, and integration interfaces.

User Documentation

Develop user manuals, FAQs, and online help resources to assist users in using the app effectively.

PLC – Execution Phase

An Example of an Execution Phase

Step 7: Training

Staff Training

Conduct training sessions for bank staff and customer service representatives.

Training includes how to use the app, how to troubleshoot common issues, and how to assist customers effectively.

Stakeholders Demonstrations

Organize demonstrations for stakeholders to showcase app functionality and features, ensuring it aligns with their expectations and gathers further feedback.

PLC – Execution Phase

Concluding Remarks of Execution Phase

The execution phase is where the planned activities are turned into tangible deliverables.

It requires effective coordination of all project components—people, processes, and technology—to ensure the mobile banking application is developed successfully, tested thoroughly, and ready for deployment.

As this phase wraps up, the project moves into the monitoring and controlling processes to ensure everything aligns with the project management plan before heading to the final closure phase.

Closure Phase

PLC – Closure Phase

Closure Phase

- This final phase ensures that the project is brought to an orderly end.
- Ensuring that all aspects of the project are completed, confirmed by the customer, and formally closed.
- Delivering the final product to the customer, and handing over project documentation.
- Evaluating what went well and what didn't, documenting lessons learned to improve future projects.
- Disbanding the project team and releasing resources back to the organization or on to other projects.

PLC – Closure Phase

An Example of Closure Phase

Closure Phase: Mobile Banking Application Development Project

Step 1: Completion and Handover

Final Testing and Approval

Conduct the final round of testing, including load testing and security testing, to ensure the app is robust and secure.

Obtain final approval from the stakeholder committee, which includes senior bank management and IT leads.

App Deployment

Deploy the mobile banking application to the production environment.

This includes making the app available for download in app stores and ensuring it integrates seamlessly with live databases and transaction systems.

Customer Rollout

Initiate a phased rollout to customers, starting with a small group to gather initial feedback and ensure smooth operation before a full rollout.

PLC – Closure Phase

An Example of Closure Phase

Step 2: Documentation and Deliverables

Project Documentation

Compile all project documentation, including design documents, test records, compliance approvals, and deployment guides, to hand over to the bank's IT operations team.

User and Maintenance Guides

Provide comprehensive user guides for customers and maintenance and support manuals for the IT operations team.

PLC – Closure Phase

An Example of Closure Phase

Step 3: Project Review

Post-Implementation Review

Conduct a meeting with the project team and key stakeholders to review the project's successes and challenges. Discuss what worked well and what could be improved for future projects.

Lessons Learned

Document lessons learned during the project, focusing on insights that can help improve future projects. This includes aspects related to project management processes, technology implementation, stakeholder engagement, and risk management.

PLC – Closure Phase

An Example of Closure Phase

Step 4: Release Resources

Resources Release

Release project resources, including staff, contracted vendors, and technologies.

Ensure that all contractual obligations are fulfilled, final payments are processed, and official contracts are closed.

Team Recognition

Recognize the efforts of the project team through formal acknowledgments, rewards, or celebration events.

This helps in team motivation and retention.

PLC – Closure Phase

An Example of Closure Phase

Step 5: Final Reporting

Final Project Report

Prepare a final project report summarizing the project outcomes, budget usage, timeline adherence, and scope achievement.

Include performance metrics and outcomes of the business objectives, like user adoption rates and operational efficiencies gained.

Stakeholder Presentation

Present the final report to senior management and other key stakeholders.

Provide insights into the project's impact on the business and potential next steps for further enhancements or additional features.

PLC – Closure Phase

An Example of Closure Phase

Step 6: Archiving

Archiving Files

Archive all project-related documents and materials in a secure environment.

This ensures that all project information is available for future reference, audits, or any subsequent projects.

PLC – Closure Phase

An Example of Closure Phase

Remarks

The closure phase formally concludes the project by ensuring all project deliverables have been satisfactorily completed and accepted.

It also involves evaluating the project process, capturing lessons learned, and effectively releasing project resources.

This phase not only wraps up the project in a structured manner but also sets a foundation for ongoing support and future initiatives.

With the successful closure of the project, the mobile banking application is now fully operational, with ongoing operations handed over to the bank's IT and operational teams.

PROJECT SCHEDULING TECHNIQUE

Introduction

Important Questions During Project Implementation

- What activities can produce expected project outputs?
- What is the sequence of these activities?
- What is the time frame for these activities?
- Who will be responsible for carrying out each activity?

Introduction

The following methods may be used to answer the above questions:

- Gantt chart
- Project Evaluation and Review Techniques (PERT)
- Critical Path Method (CPM) or Network analysis.
- Simple formats e.g., Bar charts, timelines, milestones etc.

What is Project Scheduling?

Project Scheduling:

- is the tool that communicates;
 - what work needs to be performed
 - which resources of the organization will perform the work
 - the timeframes in which that work needs to be performed.
- It reflects all of the work associated with delivering the project on time.
- Without a full and complete schedule, the project manager completes the project effort in terms of cost and resources necessary to deliver the project.

Project Scheduling ...

- A project is a collection of tasks that must be completed in minimum time or at minimal cost.
 - what work needs to be performed
 - which resources of the organization will perform the work
 - the timeframes in which that work needs to be performed.
- **Objectives of Project Scheduling**
 - Completing the project as early as possible by determining the earliest start and finish of each activity.
 - Calculating the likelihood that a project will be completed within a certain time period.
 - Finding the minimum cost schedule needed to complete the project by a certain date.

Project Scheduling ...

- Scheduling *means the estimation of time and resources required to complete activities and organize them.*
- The following tools can be used in project scheduling
 - Critical Path Method
 - PERT chart
 - Gantt chart

Network Planning & Network Diagrams

Network planning

- Network planning is the process of organizing a network to ensure efficient communication and data transfer.
- Network analysis incorporates a variety of techniques used to help plan and manage the schedule of interrelated activities.
- Network-Based Techniques are project management methods that involve representing project activities and their interdependencies using graphical diagrams.
- Useful in project management for:
 - Scheduling
 - Monitoring project
 - Controlling
 - Resource allocation
 - Reviewing costs
- Common for complex projects involving hundreds of activities

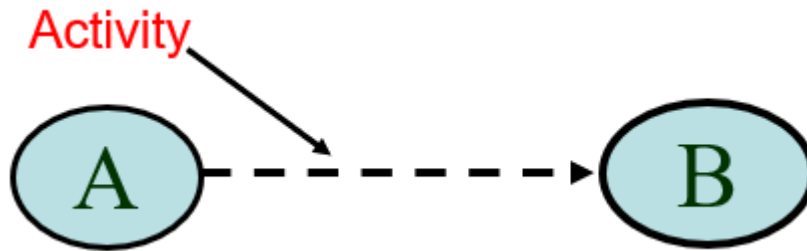
Network planning ...

- **A network diagram:** This is a diagrammatic representation of the individual project activities in their logical & practical sequence that establishes a relationship.
- **An activity:** This is the task or part of a work that requires *time* and *resources* for its implementation (the opposite is a dummy activity).
- **An event:** Starting or ending point of an activity (tail and head event)

Network planning ...

Important Network Notations

- Network is a diagrammatic illustration of the relationship btw a group of activities. A network is made of two things:
 - Activity presented as an arrow 
 - Events presented as a node 



Network planning ...

Rules for Drawing Network Diagram

1. Each project activity should be presented by **only one arrow**.
2. **An activity can begin only when all its successors are done**
3. **No two activities** can be identified by same tail and head
4. All activity flows must be shown as **moving from left to right**; so is the numbering of events.
5. Introduce as few dummy activities as may be necessary

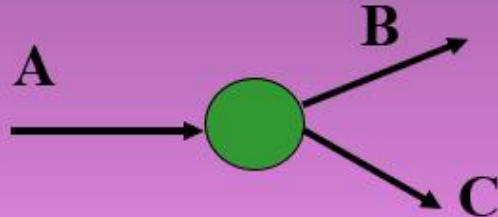
Network planning ...

Rules for Drawing Network Diagram

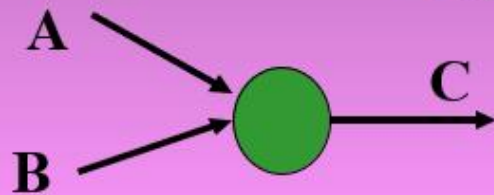
6. There should be **no circuits or loops in network** (nor circulating)
7. A network should have **only one initial and one terminal**
8. Proper activity precedence must be followed – **presented in practical and logical sequence.**
9. Network arrows are not drawn to scale but their direction is important. (**length of an arrow $\neq$ the duration of the activity**)

Network planning ...

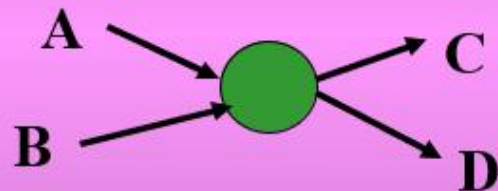
Situations in Network Diagram



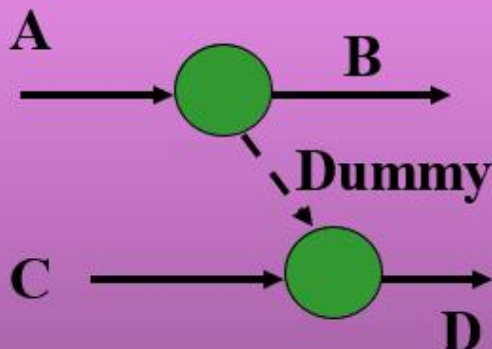
A must finish before either B or C can start



both A and B must finish before C can start



both A and B must finish before either of C or D can start



A must finish before B can start

both A and C must finish before D can start

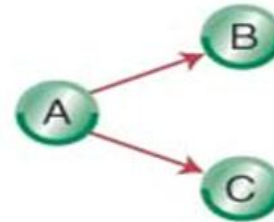
Network Diagrams

Activity-on-Node (AON)

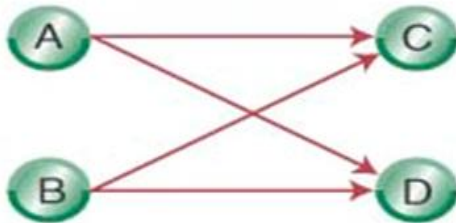
- Uses nodes to represent the activity
- Uses arrows to represent precedence relationships



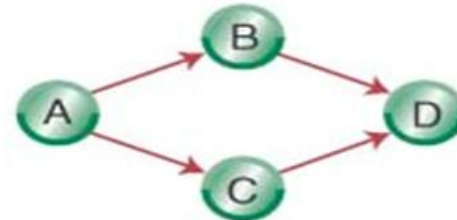
a. Activity A precedes activity B, which precedes activity C



b. Activity A must be completed before activities B and C can begin.



c. Activities A and B must both be completed before activity C or D can begin.



d. Activities B and C can begin once activity A has been completed; activity D cannot begin until both B and C are completed.

Network-Based Techniques

What is the Critical Path Method (CPM)?

- Is the network-based technique used which shows the sequence of scheduled activities that determines the duration of the project.

Or

- Is the longest sequence of tasks in a project plan that must be completed on time in order for the project to meet its deadline
- A delay in any task on the critical path, delays the whole project
- Although many projects have only one critical path, some projects may have multiple critical paths.

Network-Based Techniques

CPM ...

- The critical path is important when managing a project because it identifies all the tasks needed to complete the project
- Then determine the tasks that must be done on time, those that can be delayed if needed (due to the triple constraint of time, cost, and scope), and how much float you have.
- With this knowledge, you can prioritize, set realistic deadlines, and assign critical tasks to your most capable team members.

STEPS in CPM

CPM ...

1. Define/list all project activities
2. Estimate the activity durations.
3. Identify dependencies (interrelationships) between activities.
4. Create a network diagram.
5. Analyze the network by calculating the critical path and determining project duration. Calculate the Early Start (ES) Time , and Early Finish (EF) Time
 - ES or EST is the term used in the CPM to describe the earliest time that an activity can begin

STEPS in CPM ...

- In project management, an activity is a set of actions that are performed to produce a project's deliverables.
- Activities are the smallest parts of a project that are used for planning, tracking, and control.
- The total project is subdivided into activities and each activity is given an alphabetical symbol/code.
- This involves a detailed delineation of the activities to be performed to complete the project.

STEPS in CPM ...

Step1: Prepare the list of activities

- Take an example of activities in Table 01.

Table 01: List of activities

S/N	Activity	Symbol / Code
1	Site Selection	A
2	Digging well	B
3	Laying field channels	C
4	Procurement of pump	D
5	Installation of pump	E
6	Test run	F

STEPS in CPM ...

Step 2: Define the interrelationship among the activities (activity Sequence).

- The relationship among the activities could be defined by specifying the preceding and succeeding activity.
- The preceding activity for an activity is its immediate predecessor, i.e. the activity that needs to be completed before the start of the new activity.

Table 2: Activity Sequence

S/N	Activity	Symbol/code	Preceding activity
1	Site selection	A	-----
2	Digging well	B	A
3	Laying field channels	C	B
4	Procurement of pump	D	A
5	Installation of pump	E	B, D
6	Test runs	F	C, E

STEPS in CPM ...

Step 2: Define the interrelationship among the activities

- In the given example, “the selection of the site” precedes the digging of the well.
- In other words, “the site” needs to be selected before digging the well. Thus the activity “Selection of the site” becomes the proceeding activity to the activity of “Digging the well”
- Succeeding activity is the one that immediately starts after completion of the activity. “**Digging well**” is the succeeding activity to “**Selection of the site**”.

STEPS in CPM ...

Step 3: Estimate Activity Duration

- The activity time is the time that is actually expected to be expended in carrying out the activity.

Table 03: Activity Time Estimates

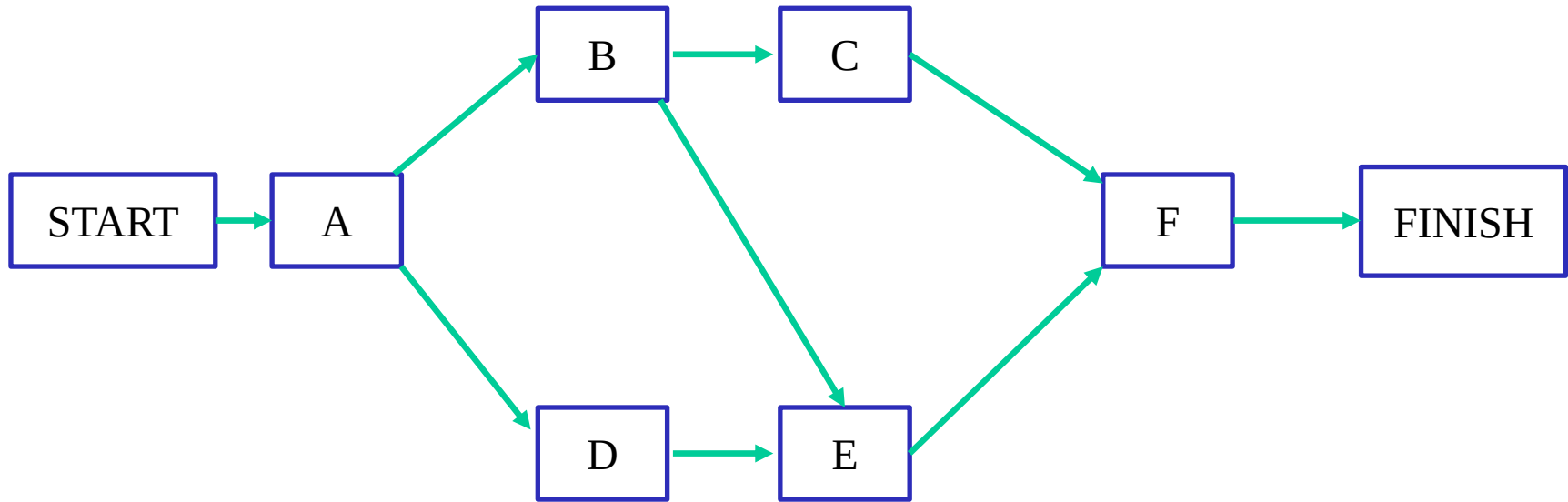
S/N	Activity	Symbol/Code	Preceding Activity	Estimated Time
1	Site selection	A	-----	7
2	Digging well	B	A	3
3	Laying field channels	C	B	15
4	Procurement of pump	D	A	7
5	Installation of pump	E	B, D	3
6	Test run	F	C, E	2

STEPS in CPM ...

Step 4: Draw the Network Diagram

- The activities identified and their dependencies aid you in drawing the critical path analysis chart (CPA)
- Is also known as the network diagram.
- The network diagram is a visual representation of the order of your activities based on dependencies.

Step 4: Draw the Network Diagram ...

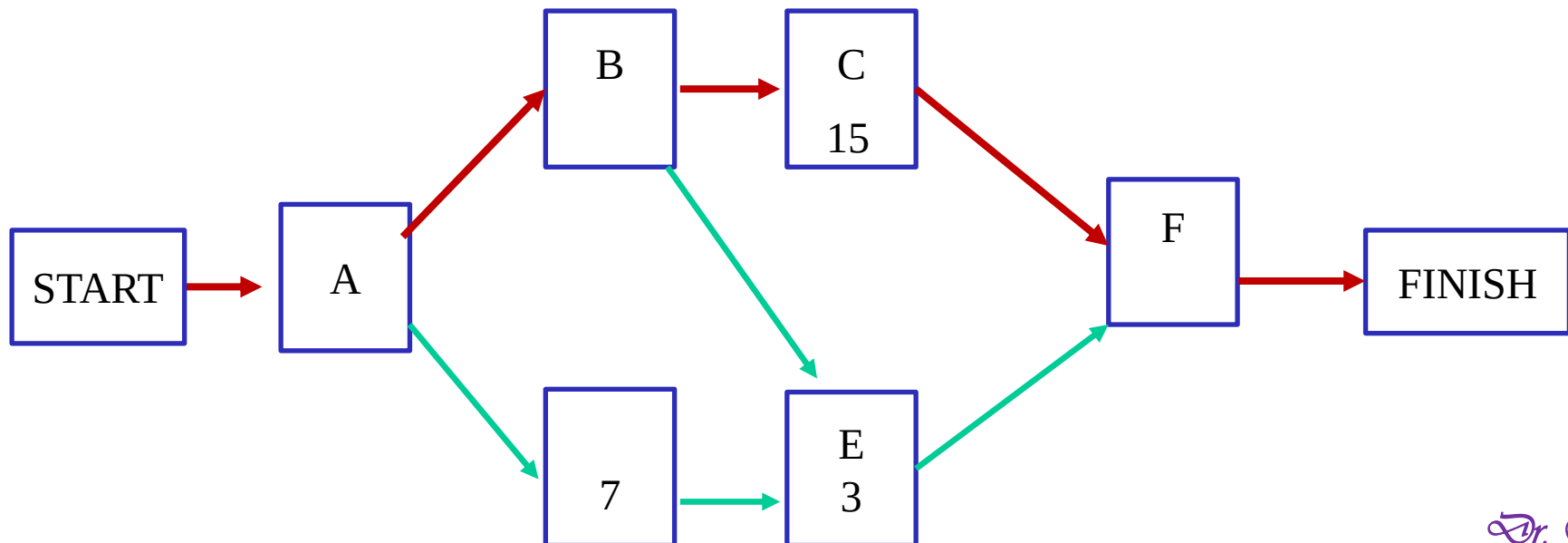


STEPS in CPM ...

Step 5: Identify the Critical Path

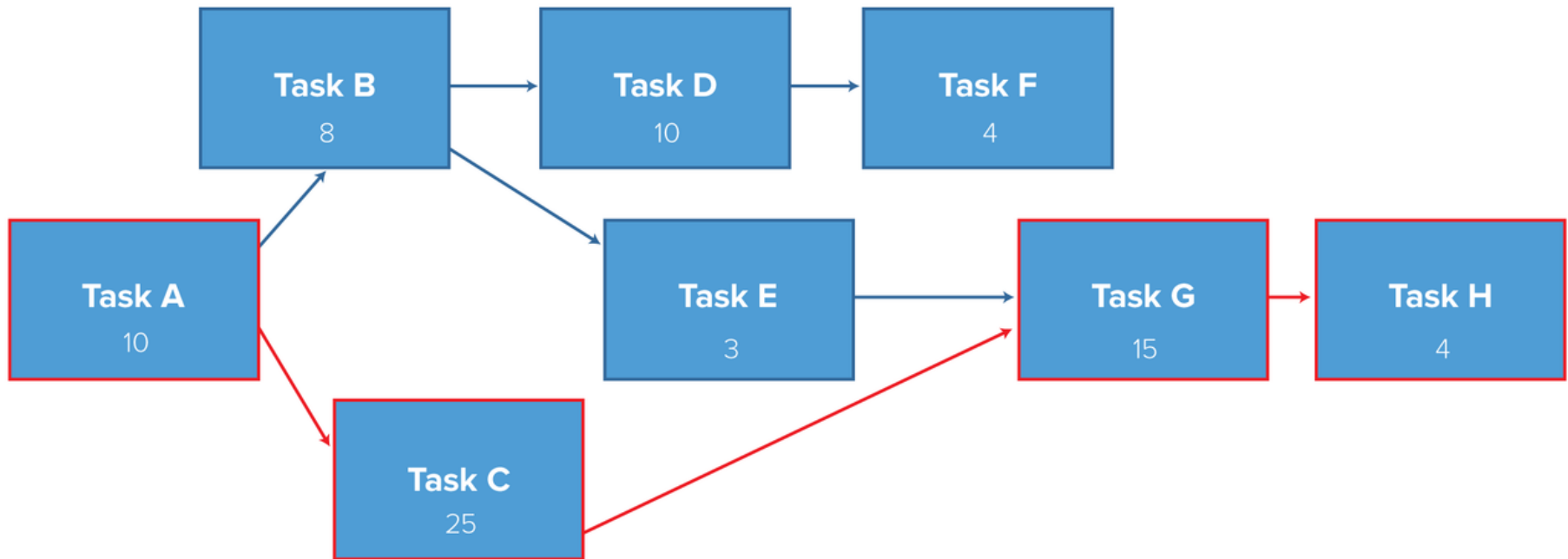
Two ways you can now identify the critical path

1. Eyeball the network diagram and simply identify the longest path throughout the network: - the longest sequence of activities on the path.
- Look for the longest path in terms of the longest duration in days, not the path with the most boxes or nodes



STEPS in CPM ...

Step 5: Identify the Critical Path



STEPS in CPM ...

Step 5: Identify the Critical Path

Two ways you can now identify the critical path

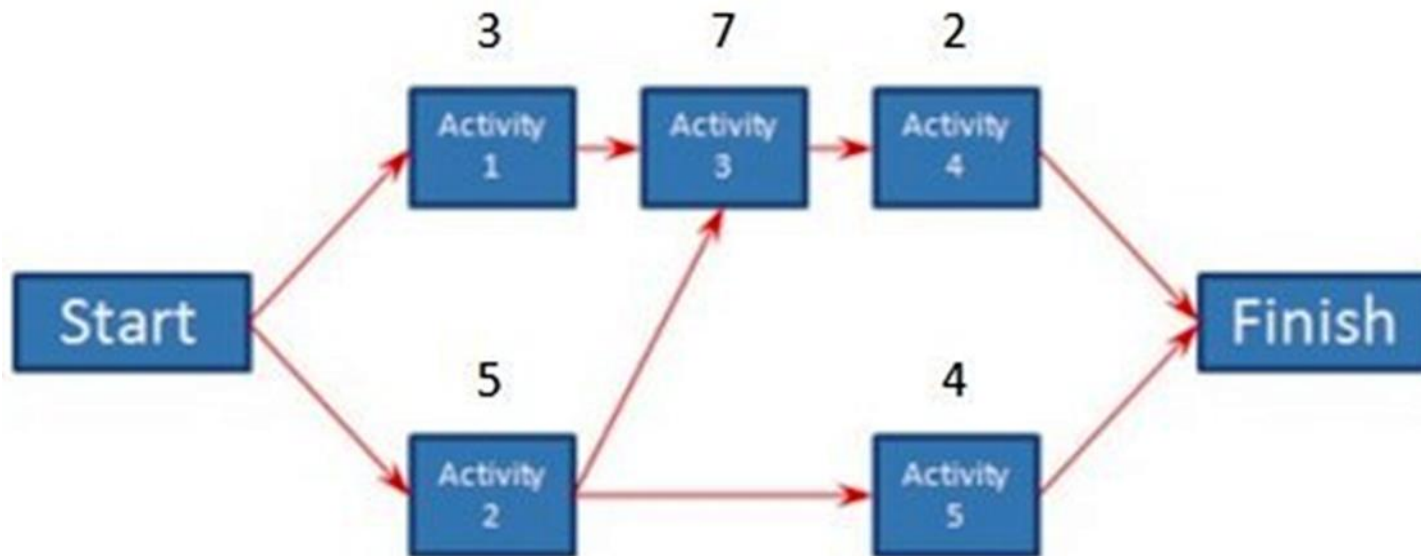
2. Secondly Identify critical activities with the Forward Pass/Backward Pass technique.
 - Enable you to identify the earliest start (ES) and Earliest finish(EF) times, and the latest start (LS) and (LF) finish times for each activity

Importantly Things in CPM

- The CPM has four key elements
 - Critical Path Analysis
 - Float determination
 - Early Start & Early Finish Calculation
 - Late Start & Late Fish Calculation
- CP is the sequence of activities with the longest duration
- Delay in one activity leads to a delay in the whole project

Importantly Things in CPM

- Lets look at some critical path examples to bring out the concepts
- Suppose a project has the following activities.



- Each activities duration is above each node

Importantly Things in CPM

- To get the critical path we add the duration of each node to determine its total duration
- CP is the one with the longest duration
- In the project above there are 3 paths thru the project



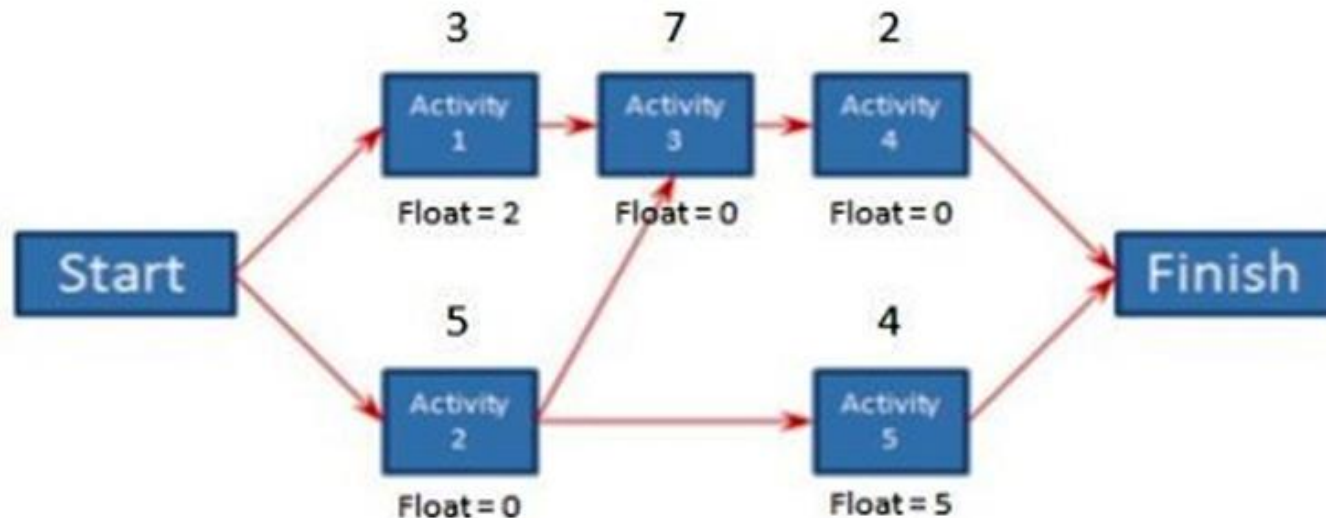
Start → Activity 1 → Activity 3 → Activity 4 → Finish	$3 + 7 + 2 = 12$
Start → Activity 2 → Activity 3 → Activity 4 → Finish	$5 + 7 + 2 = 14$
Start → Activity 2 → Activity 5 → Finish	$5 + 4 = 9$

- The Critical one is path 2

Determining the Float

Determining the Float

- Float is the amount of time (days) an activity can slip before it causes the project to be delayed
- This is sometimes called Slack
- Start with activities on the Critical Path.
- Each of those activities has a Float of 0, meaning if any of the activities slip, the project is delayed

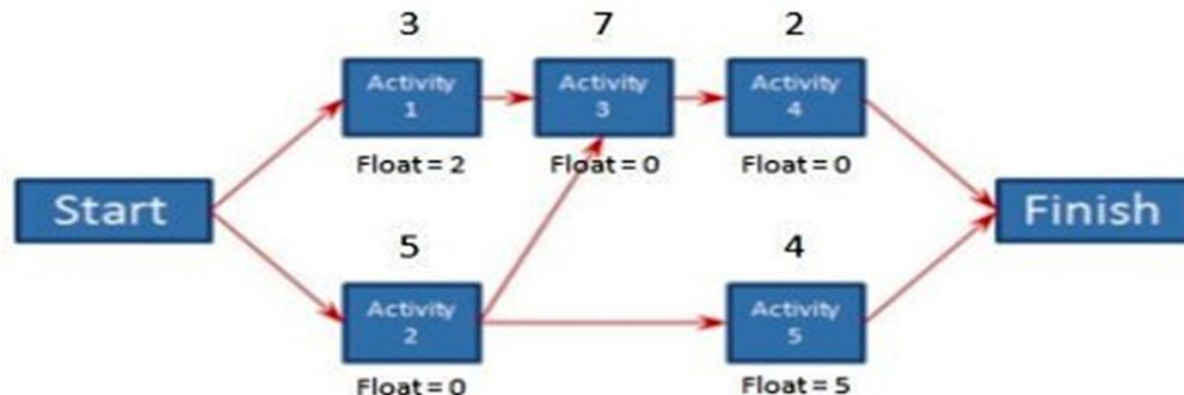


Determining the Float ...

- Take the next longest path - Subtract its duration from the duration of the critical path to get float for each of those activities
- Continue doing the same for each subsequent longest path till each activity float is determined
- If an activity is on two paths, its float will be based on the long path that it belongs to.
- Float Can also be computed as:
Total float = TF = Latest start – Earliest Start

Determining the Float ...

- In the case of our CP diagram Activities 2,3 and 4 are on a critical path so have float 0.
- The next longest path is activities 1,3 & 4. Since Activities 3 & 4 are on CP their float remains 0.
- For activity 1 float will be Duration of Critical path - duration of this path = **14 -12 =2**.
- The next path is activities 2 & 5. Activity 2 is on CP so float =0
- Activity 5 has a float of **14-9 =5**.
- Means so long Activity 5 doesn't slip more than 5 days it won't cause a delay in the project.



EARLY START & EARLY FINISH

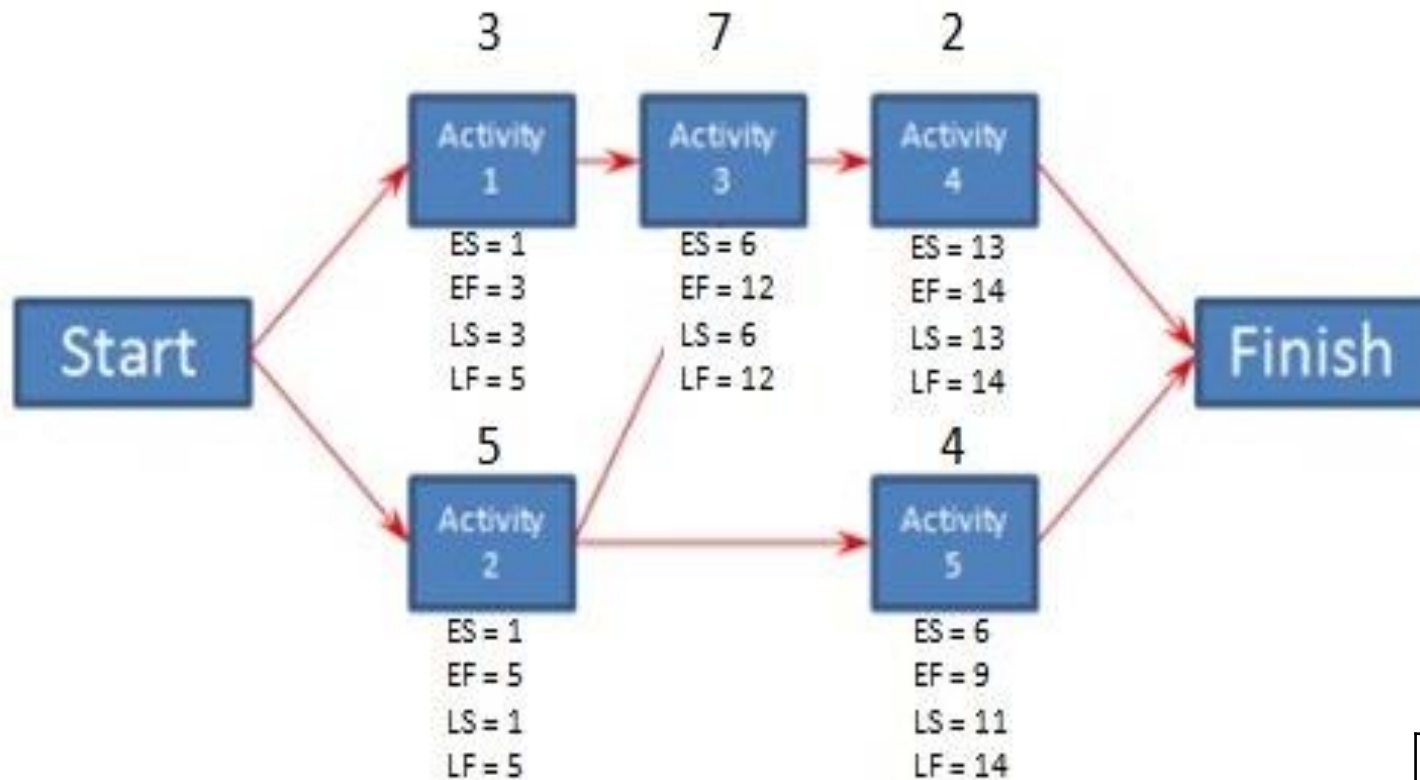
Early Start & Early Finish

- CPM includes a technique called **Forward pass**
- Used to determine *the earliest date an activity can start and the earliest date it can finish.*
- These dates are valid as long as all prior activities in that path started on their earliest start date and didn't slip.
- Starting with CP the Early Start (ES) of Activity 1 = 1
- The early finish (EF) of an activity is its ES plus its duration minus one.

Early Start & Early Finish

- Using our earlier example, Activity 2 is the first activity on the critical path:
 - $ES = 1$,
 - $EF = 1 + 5 - 1 = 5$.
- Activity 3 its ES is Previous activities $EF+1$
- Activity 3 $ES = 5 + 1 = 6$
- Its EF is calculated as before;
 - $EF = 6 + 7 - 1 = 12$
- To calculate the ES of an Activity with more than one predecessor, you will use the activity with the latest EF.

Early Start & Early Finish



ES	ACT	EF
LS	DUR	LF

LATE START & LATE FINISH

Late Start & Late Finish

- The **Backward Pass** is a CPM technique you can use to determine the latest date an activity can start and the latest date it can finish before it delays the project .
 - Start with CP but begin from last activity of the path
 - The **Late Finish (LF)** for the last activity in every path is the same as the last activity's EF in the critical path.
 - The **Late Start (LS)** is the $LF - \text{duration} + 1$.
 - For E.g. Activity 4 is the last activity on the critical path. Its **LF is the same as its EF, which is 14.**
 - To calculate **the LS**, subtract its duration from its LF and add one. $LS = 14 - 2 + 1 = 13$.

Late Start & Late Finish

- The next Activity in the critical path is Activity 3.
- Its LF is equal to Activity 4 LS - 1.
- Activity 3 LF = $13 - 1 = 12$.
- It's LS is calculated the same as before by subtracting its duration from the LF and adding one.
- Activity 3 LS = $12 - 7 + 1 = 6$.
- CPM is an important tool for managing project schedules
- With large projects with many activities PM software easily does this quickly

Summary

- **Free slack** or **free float** is the amount of time an activity can be delayed without delaying the early start of any immediately following activities
- **Total slack** or **total float** is the amount of time an activity may be delayed from its early start without delaying the planned project finish date
- A **forward pass** through the network diagram determines the early start and finish dates
- A **backward pass** determines the late start and finish dates

Differences between PERT & CPM

BASIS	PERT	CPM
Meaning	PERT is a Project management technique, used to manage uncertain activities of a project	CPM is a statistical technique of project management that manages well defined Activities of a project.
What is it?	A technique of planning and control of time	A method to control cost and time.
Orientation	Event-oriented	Activity-oriented
Usage	Used in Research & Development project – activities are unpredictable	Used in Construction project – Activities are Predictable
Model	Probabilistic Model – cant estimate completion time	Deterministic Model
Focuses on	Time - meeting time target	Time-cost trade-off – optimize cost min time duration
Estimates	Three time estimates -Optimistic, most likely & pessimistic time	One time estimate
Suitable for	projects needing High precision time estimate	Projects needing Reasonable time estimate
Management of	Unpredictable Activities	Predictable activities
Nature of jobs	Non-repetitive nature	Repetitive nature